CROSSCODER MODEL DIFFING

FROM MODEL DIFFS TO CAUSAL CONTROL

A shared sparse representation for explaining and testing behavioral differences between code models

BEHAVIOR

↓

FEATURES

↓

STEERING

RESEARCH QUESTION

Evaluation tells us what changed. CrossCoder asks which internal differences can explain—and control—it.

BEHAVIOR

Which tasks changed?

pass → fail
fail → pass

What failure mode changed?

REPRESENTATION

Paired layer-16 residuals

Same-text alignment

Shared sparse features

CAUSAL TEST

Choose candidate directions

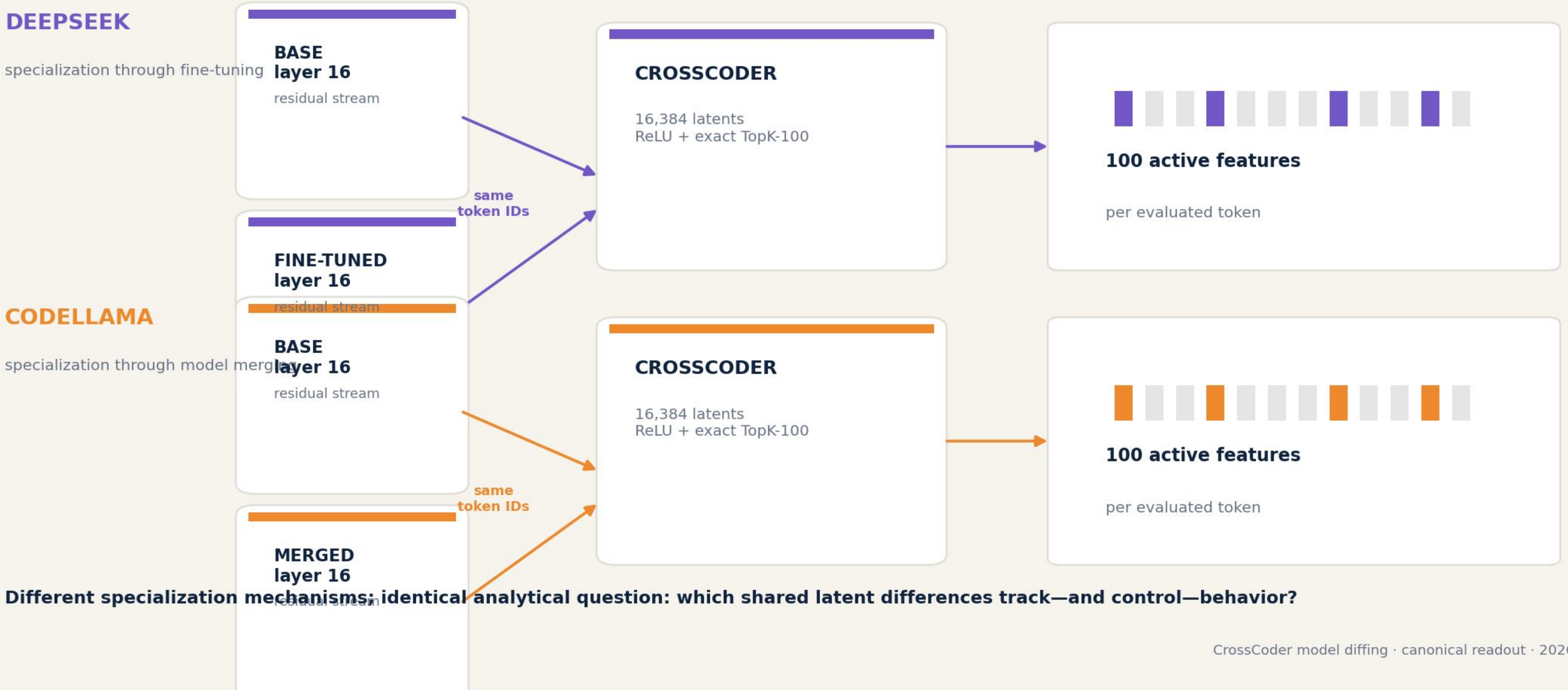
Intervene during generation

Compare against controls

Behavior localizes the model difference. Sparse features turn it into a testable hypothesis.

ONE ANALYSIS FRAMEWORK, TWO VERY DIFFERENT MODEL CHANGES

Same token IDs turn each model pair into a shared sparse vocabulary

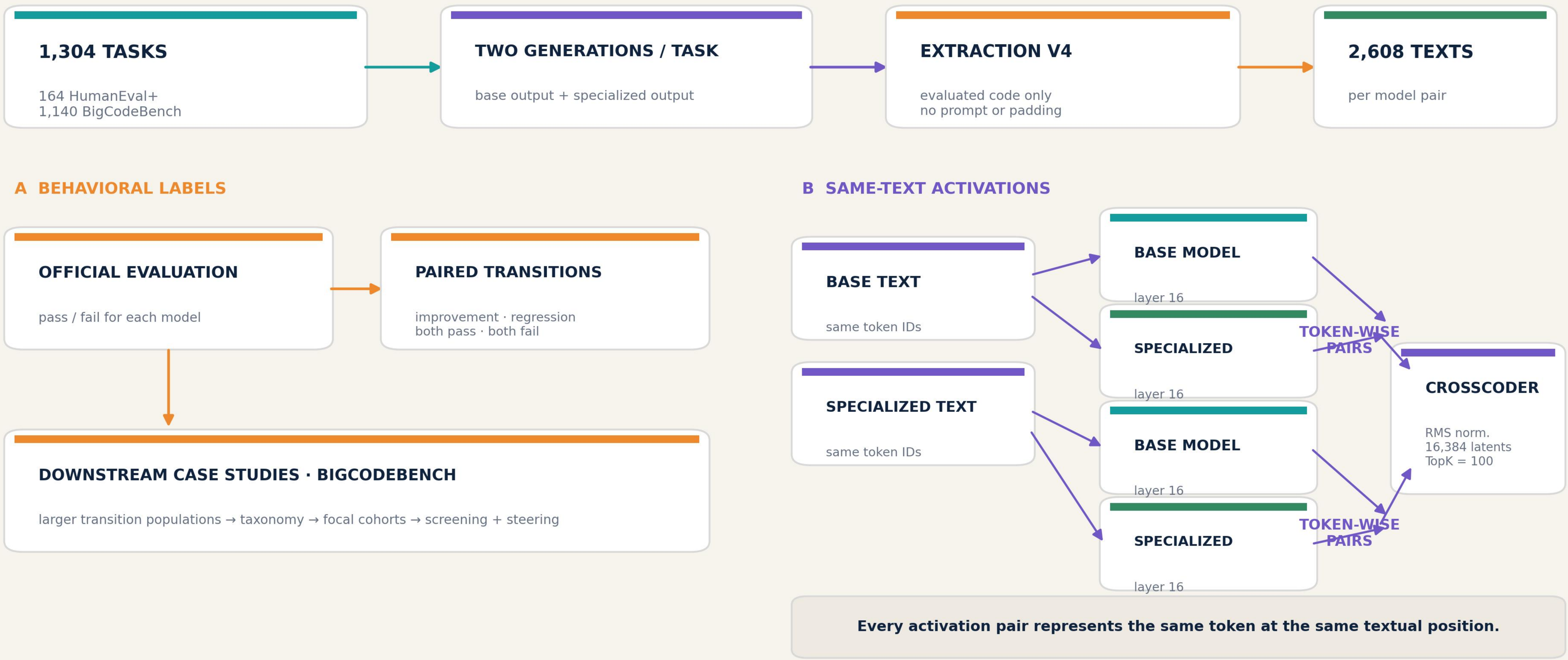


Different specialization mechanisms; identical analytical question: which shared latent differences track—and control—behavior?

SAME-TEXT CROSSCODER DATASET

FROM BENCHMARK GENERATIONS TO A SHARED SPARSE REPRESENTATION

Evaluation and same-text replay reuse the same extracted code for different purposes



THE TWO SPECIALIZATIONS MOVE PERFORMANCE IN OPPOSITE DIRE

Canonical extraction-v4 labels · same raw generations, audited reprocessing

DEEPSEEK · BIGCODEBENCH

268/1140
BASE

404/1140
FINE-TUNED

+11.93 pp

BigCodeBench
pass-rate change

CODELLAMA · BIGCODEBENCH

314/1140
BASE

27/1140
MERGED

-25.18 pp

BigCodeBench
pass-rate change

DeepSeek: 215 improvements · 79 regressions

CodeLlama: 4 improvements · 291 regressions

RECATCHER CONTEXT

Directionally consistent motivation: specialization can improve aggregate capability while still creating localized regressions.

Not a direct numerical replication: ReCatcher used 10 generations/prompt and different regression definitions.

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination
48: wrong output or logic
16: missing name/import
32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch
50: wrong logic/other runtime
43: edge case/exception
78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases
Improvements: only 4 positives · EXCLUDED

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and oft

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL
Correct-looking body continues into:
`#tests/test_task_func.py`
`import pytest`

FINE-TUNED · PASS
Completes the function and stops after a short example.

First textual divergence ≈76% of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS
parses XML → writes JSON file → returns result

MERGED · FAIL
`return xmltodict.parse(s)`
omits the required file write side effect

First divergence ≈81% of base code;
target region is the late function body.

FEATURE SCREENING

We rank stable behavioral contrasts—not raw activation alone

4 SUMMARIES

max · early_max
mean · active fraction

→

Δ PER TASK

aggregate(specialized text)
− aggregate(base text)

→

OBSERVED EFFECT

mean Δ positives
− mean Δ controls

→

200 PERMUTATIONS

shuffle labels
recompute effect

→

E/V

effect ÷ null SD

Minimum support

max(3 tasks, 10% of positives)

Interpretation

E/V is permutation signal-to-noise—not a z-score or p-value.

Positive and negative orientations are retained: either enrichment direction may support behavior-aligned steering.

ERROR-FOCUSED FEATURE SELECTION

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%

16,384 → **≈30–40** → **≈0.2%**

latents / CrossCoder — candidates / model — of latent space

WHY |α| = 3?

Earlier dose-response work identified a practical exploratory point: visible causal changes below the most disruptive high-dose magnitude.

WHY STANDARDIZE?

One common magnitude makes dozens of candidates comparable before selecting five features for com

|α|=3 is a common operating point—not an independently validated optimum.

WHAT THE SCREENING ACTUALLY RETURNS

Representative valid cells; E/V is the primary ranking signal

DEEPSEEK · IMPROVEMENT ASSOCIATION

Rank	Feature	Effect	E/V
1	3602	-0.021	**-8.23**
2	10168	-0.018	**-7.67**
3	5422	-0.005	**-7.66**
4	1982	-0.015	**-7.53**
5	1862	-0.008	**-7.45**
6	1650	-0.026	**-7.28**
7	5039	-0.088	**-7.26**
8	9580	-0.013	**-7.24**
9	5837	-0.010	**-7.18**
10	9537	-0.038	**-7.15**

CODELLAMA · REGRESSION ASSOCIATION

Rank	Feature	Effect	E/V
1	13147	+0.126	**+5.24**
2	8992	-0.011	**-4.93**
3	1840	-0.046	**-4.77**
4	16161	+0.127	**+4.76**
5	15471	+0.005	**+4.75**
6	10570	+0.188	**+4.60**
7	8138	-4.973	**-4.51**
8	7353	+0.228	**+4.48**
9	14359	-0.166	**-4.46**
10	2825	+0.338	**+4.38**

Examples—not a single global ranking. Candidate pools are formed by union and deduplication across valid cells.

FROM BEHAVIORAL TRANSITIONS TO ERROR-FOCUSED SCREENING

ERROR-FOCUSED TOP-10 LISTS DEFINE THE NEW α=3 CANDIDATE POOL

Three summaries per cell: max · mean · active fraction; ranking by absolute E/V

DEEPSEEK · 80 CONTAMINATION IMPROVEMENTS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13879	14175	**2913**	2468
6895	9666	**7728**	14295
1078	1206	15821	8661
596	8686	12308	**7728**
8185	7169	16264	**2913**
14820	11612	**2957**	**2957**
1939	7804	**3602**	9700
15235	**1939**	2621	6305
15733	4994	6877	10736
16205	1591	14006	**3602**

40 nominations → 35 unique features

5 feature(s) appear in more than one list

CODELLAMA · 50 WRONG-LOGIC REGRESSIONS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
7692	**15559**	8875	**4173**
10818	3097	**7588**	**15559**
5642	565	12895	12623
447	**4173**	**447**	15968
11596	15890	**4634**	8142
1017	**414**	5862	9419
4634	4740	3366	13231
7588	13768	7486	**414**
3442	4309	**1017**	13151
8384	14969	**5642**	13172

40 nominations → 32 unique features

8 feature(s) appear in more than one list

NEW α=3 SWEEP: exactly 35 DeepSeek + 32 CodeLlama unique features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.

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candidates / model

of latent space

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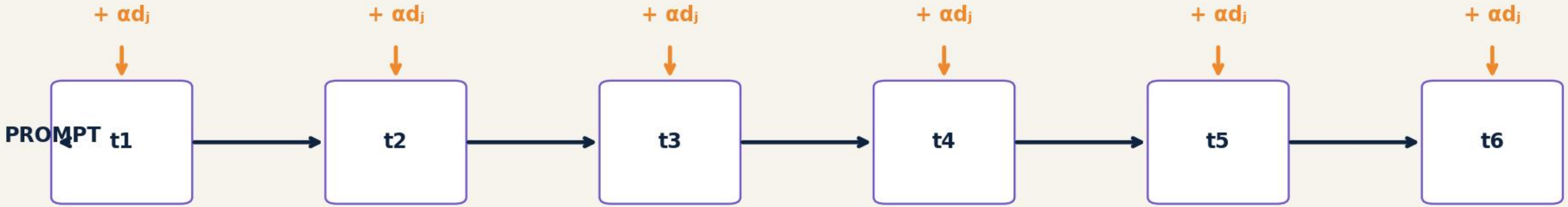
OUR CURRENT INTERVENTION IS CONTINUOUS STEERING

The selected decoder direction is added at every autoregressive generation step



CURRENT PROTOCOL

Layer 16 · last residual position
feature decoder on intervened model side
applied through the full generation

WHAT IT IS NOT

Not a one-token intervention
not activation-gated or clamped
not conditional on natural activation

This tests continuous control of the generation trajectory—not whether one naturally active token alone caused the error.

THE COMPLETE RESULT SEPARATES A STRONGER AND A WEAKER REGI

Official BigCodeBench 0.1.5 evaluation · α=3 · continuous last-token steering

DEEPSEEK

Features evaluated	35/35
Baseline	0/80
Feature-task transitions	116
Unique corrected tasks	16/80

Several features produce repeated official corrections.

CODELLAMA

Features evaluated	32/32
Baseline	0/50
Feature-task transitions	6
Unique corrected tasks	4/50

Single-feature corrections remain sparse.

STRONGER / LOCALIZED

DeepSeek contains several multi-task causal candidates.

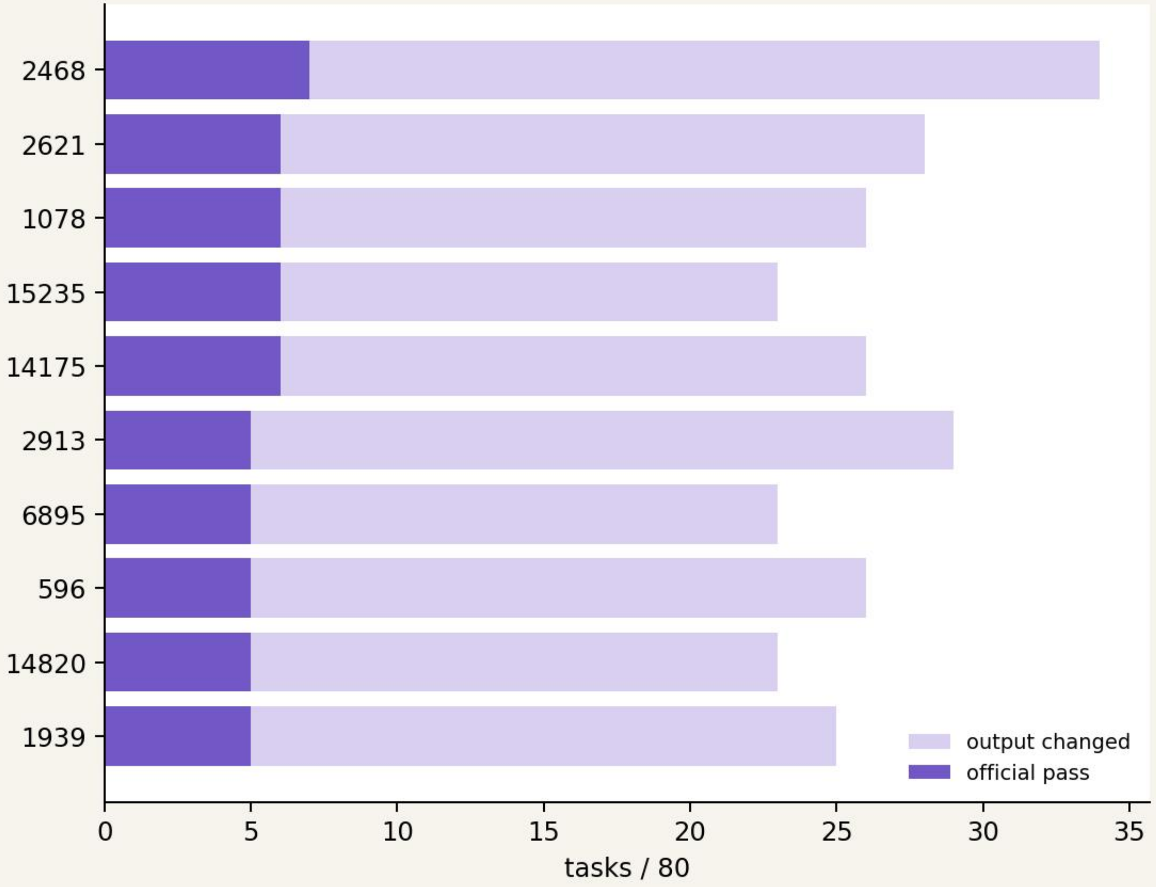
WEAKER / DIFFUSE

CodeLlama yields sparse, task-specific corrections.

COMPLETE · DeepSeek 35/35 and CodeLlama 32/32; canonical α=0 gates passed before steering.

DEEPSEEK: THE COMPLETE α=3 RANKING ADDS TWO TOP-5 FEATURES

Causal outcome first; |E/V| breaks ties among features with equal official passes



FINAL TOP 5

1. 2468   7/80 pass   34/80 changed   E/V -7.29
2. 2621   6/80 pass   28/80 changed   E/V -10.53
3. 1078   6/80 pass   26/80 changed   E/V +5.72
4. 15235   6/80 pass   23/80 changed   E/V -4.61
5. 14175   6/80 pass   26/80 changed   E/V +4.09

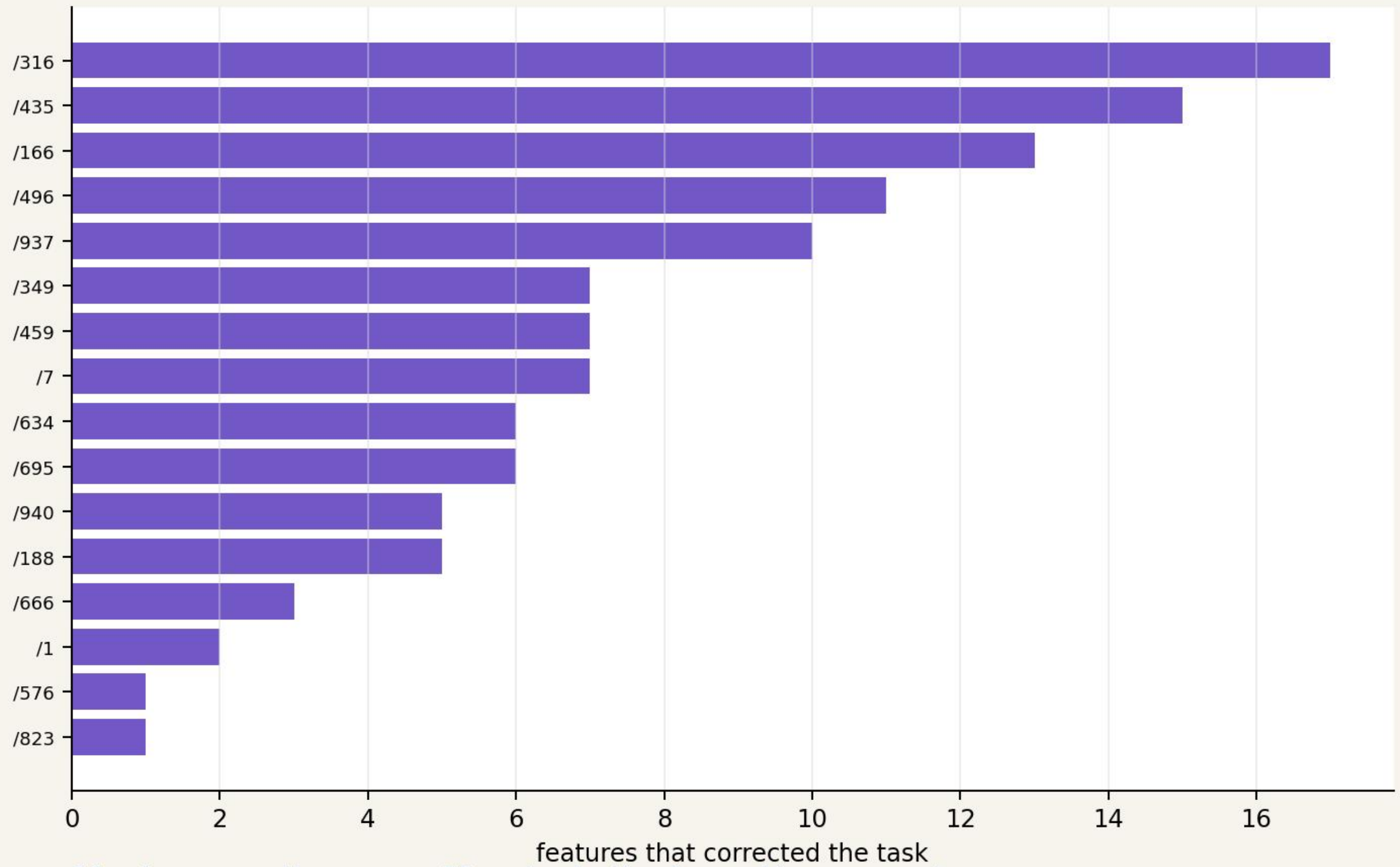
1078 and 14175 enter after completion of all 35 arms.

2913 is #6 at α=3, but later peaks at 11/80 at α=−5.

CANONICAL ERROR-FOCUSED α=3 SWEEP · COMPLETE READOUT

DEEPSEEK: 116 TRANSITIONS COLLAPSE TO 16 SUSCEPTIBLE TASKS

Feature-task outcomes are not independent causal successes



16 unique corrections across 80 contamination-focused improvements.

CONCENTRATION

/316 17 features
/435 15 features
/166 13 features
/496 11 features
/937 10 features

INTERPRETATION

The sweep finds causal levers, but a small set of tasks is broadly steering-

CANONICAL ERROR-FOCUSED α=3 SWEEP · PARTIAL READOUT

FEATURE 2468 IS THE LEADING DEEPSEEK CANDIDATE

Meaning ↔ failure mode ↔ activation timing must remain distinct

SCREENING

paired local · rank 1
E/V -7.29
summary: max
selected around the late divergence window

CAUSAL OUTCOME

7/80 official passes
34/80 outputs changed
13/80 contamination cleanups
baseline: 0/80

CAUSED TIMING

Among changed outputs:
mean first divergence: 42.7%
median: 40.8%

Continuous steering can redirect earlier than the natural activation windo

SUPPORTED local screening nominated a feature with repeated official causal effects.

NOT YET SUPPORTED feature-specificity versus sham/matched latent controls.

CODELLAMA: SINGLE-FEATURE CONTROL IS WEAK AND TASK-SPECIFIC

Complete 32-feature sweep; baseline 0/50

SIX FEATURES · ONE PASS EACH

7692 → BigCodeBench/119
10818 → BigCodeBench/119
5642 → BigCodeBench/630
11596 → BigCodeBench/119
4309 → BigCodeBench/490
8142 → BigCodeBench/823

FOUR UNIQUE TASKS

/119 corrected by 3 features
/823 corrected by 1
/490 corrected by 1
/630 corrected by 1

26/32 features produced no official correction.

Trajectory changes occur earlier and more diffusely than in the contamination-focused DeepSeek cohort.

Boundary condition: correlation and trajectory control do not imply that one latent can repair a heterogeneous merged-model regression.

CANONICAL ERROR-FOCUSED α=3 SWEEP · PARTIAL READOUT

NEXT TARGETS COMBINE CAUSAL EFFECT WITH SCREENING STRENGTH

Ordering: official passes first; |E/V| breaks ties

DEEPSEEK

RANK	FEATURE	PASS	BEST SCREEN	|E/V|
1	**2468**	**7/80**	**paired local · #1**	**7.29**
2	2621	6/80	paired global · #8	10.53
3	15235	6/80	association global · #8	4.61
4	14175	6/80	association local · #1	4.09
5	2913	5/80	paired global · #1	10.95

CODELLAMA

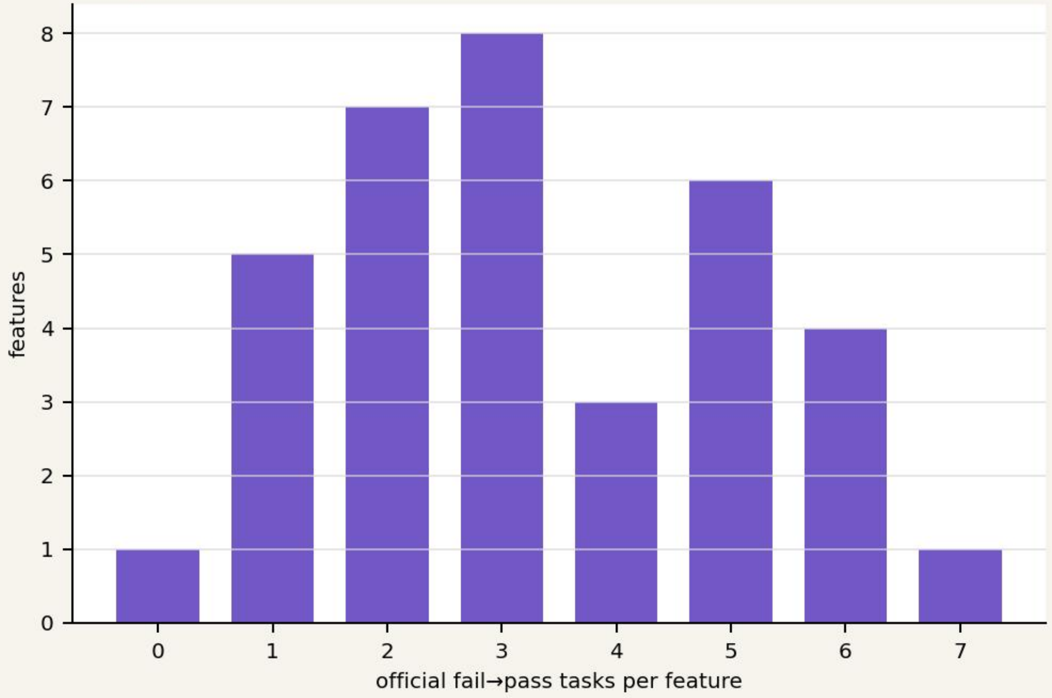
RANK	FEATURE	PASS	BEST SCREEN	|E/V|
1	**7692**	**1/50**	**association global · #1**	**5.90**
2	10818	1/50	association global · #2	5.90
3	5642	1/50	association global · #3	5.74
4	11596	1/50	association global · #5	5.61
5	4309	1/50	association local · #9	4.86

These are candidates for full dose curves plus sham and randomly selected/matched latent controls—not confirmatory winners yet.

THE α=3 SWEEP REVEALS BOTH MODEL AND TASK SENSITIVITY

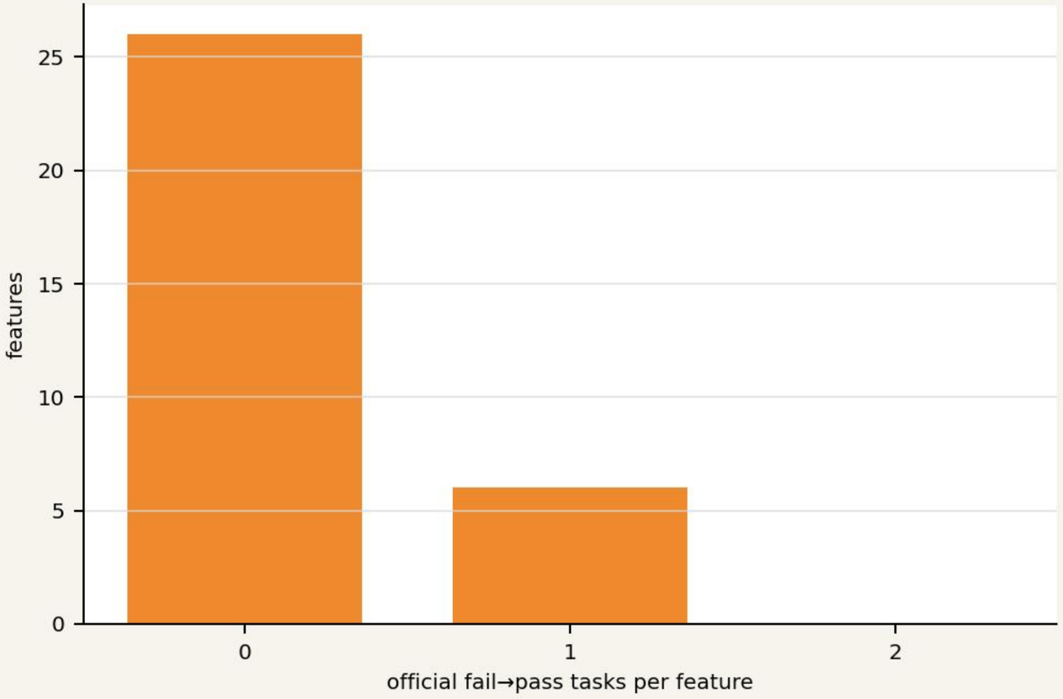
Complete error-focused sweep · official BigCodeBench 0.1.5 evaluation

DeepSeek · 35 screened features



116 feature–task transitions → 16/80 unique tasks

CodeLlama · 32 screened features



6 feature–task transitions → 4/50 unique tasks

The sweep defines five target features per model for prospective dose-response comparison with random CrossCoder features.

CONTROLLED CAUSAL DESIGN

FROM SCREENING TO A PROSPECTIVE TARGET–RANDOM TEST

Every arm shares the same tasks, canonical seed rule, generation settings, and evaluator

1 α=0 GATE

byte-exact canonical reproduction

2 TARGETS

five screen-selected features

3 RANDOM

three prospectively sampled CrossCoder latents

4 DOSE

$|\alpha| = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5$

5 ENDPOINTS

code changed + official fail→pass

INTERVENTION

Layer 16 · decoder direction on the intervened model side · last residual position · added at every autoregressive generation step · temperature 0.2 · top-p 0.95 · max 512

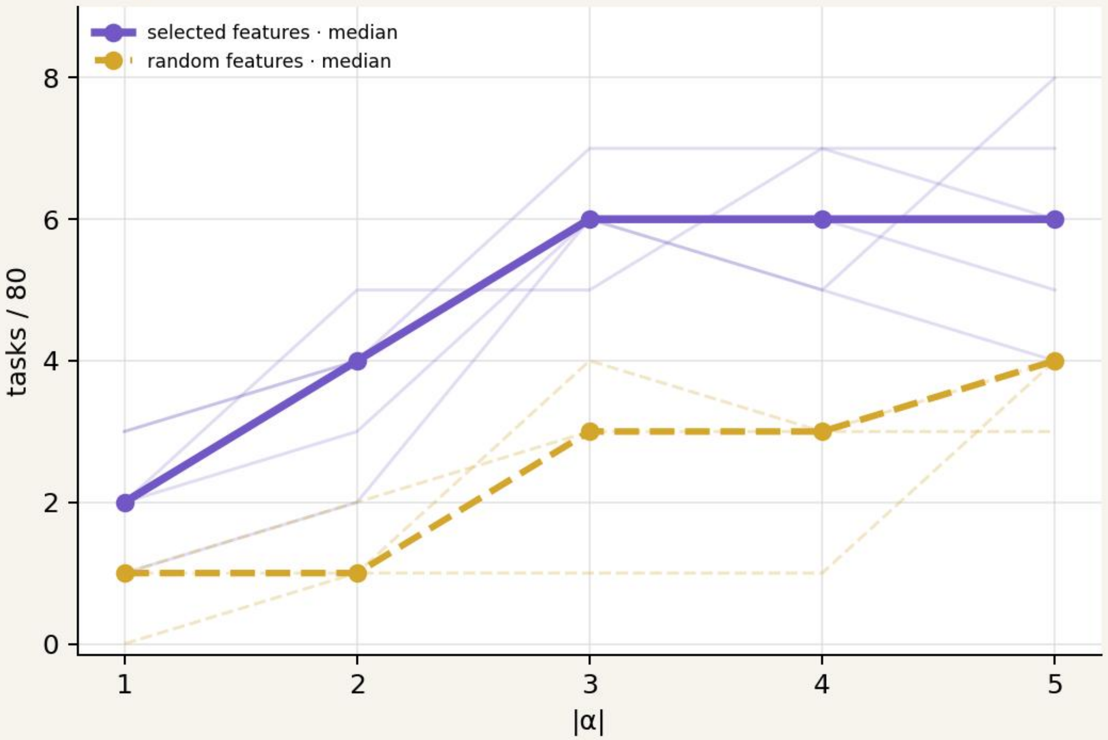
Reproduction gate passed: DeepSeek 80/80 and CodeLlama 50/50 raw completions matched byte for byte; both baselines score 0 passes.

PRIMARY COMPARATIVE RESULT

SCREEN-SELECTED FEATURES OUTPERFORM RANDOM LATENTS

Official fail→pass outcomes across the complete dose range

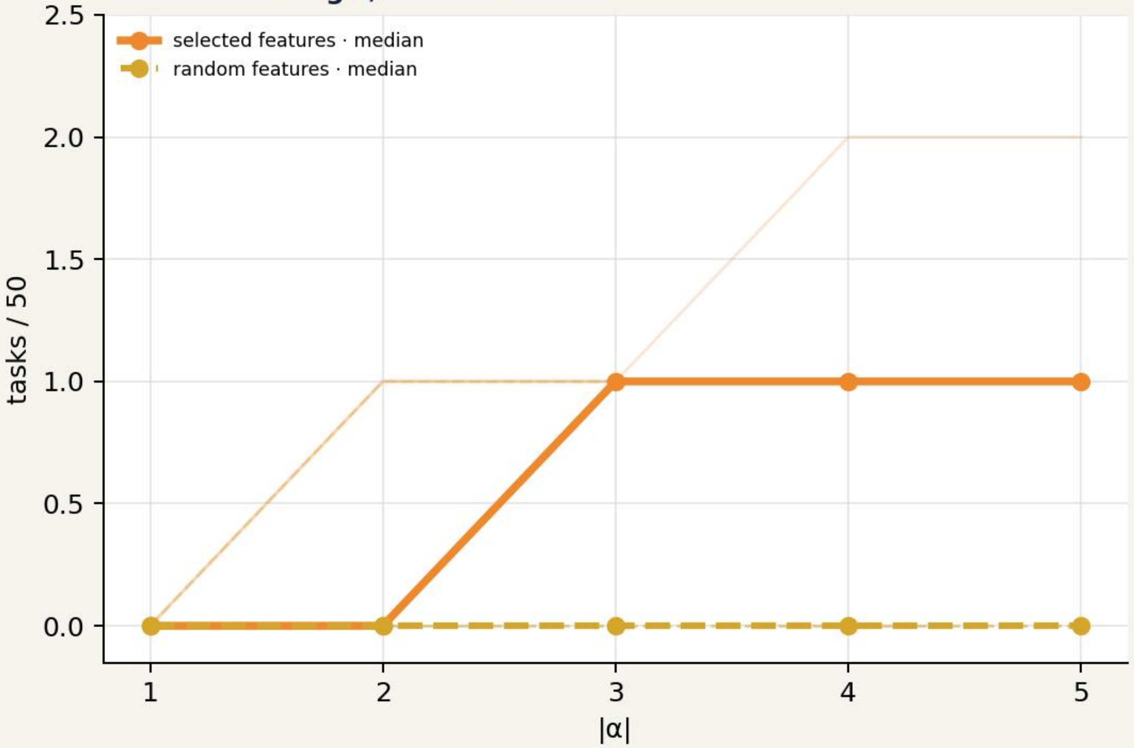
DeepSeek · contamination cohort



Selected: 16 unique tasks

Random: 10 · all overlap selected successes

CodeLlama · logic/runtime cohort



Selected: 5 unique tasks

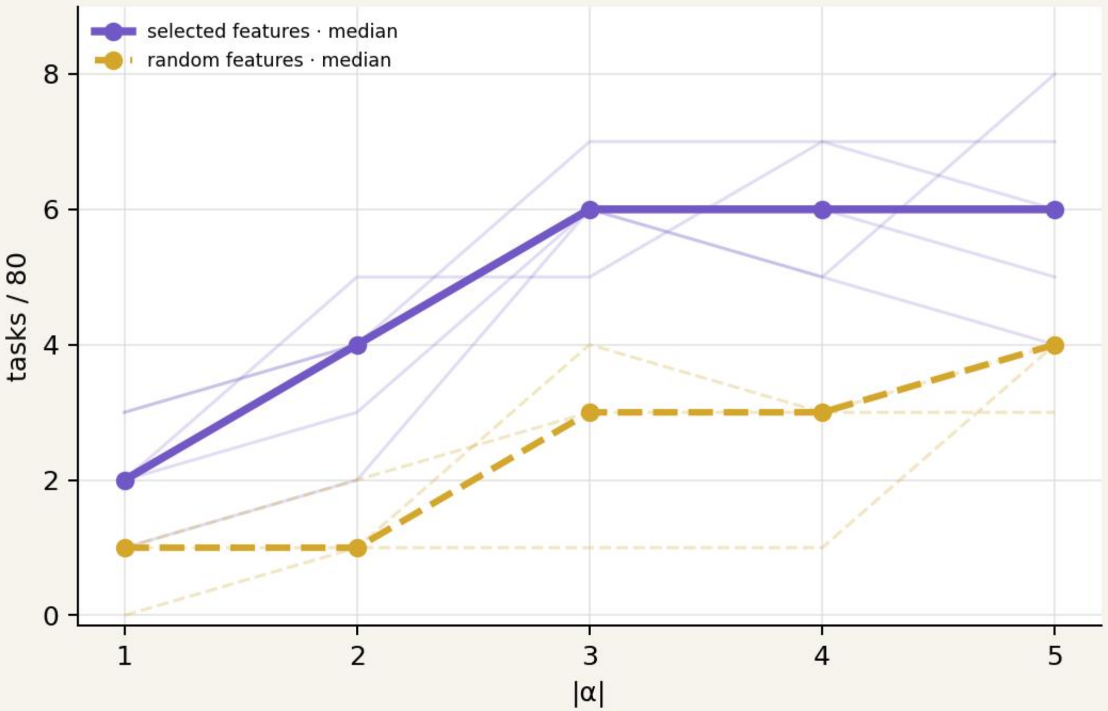
Random: 1 · overlaps a selected success

The selected pools reach six additional DeepSeek tasks and four additional CodeLlama tasks beyond the random-feature pools.

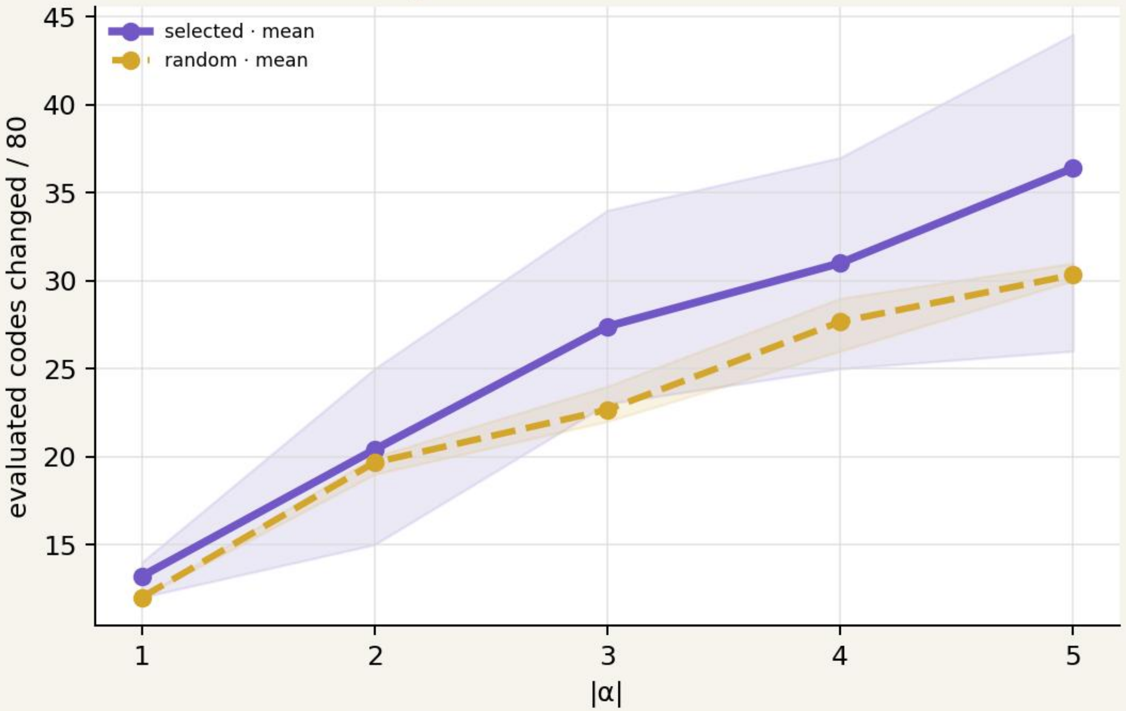
DEEPSEEK: THE SELECTED POOL SHOWS STRONGER AND BROADER CONTR

Five target features versus three prospectively sampled random CrossCoder features

Official corrections



Evaluated-code changes



OUTCOME

16 vs 10 unique tasks

SEPARATION

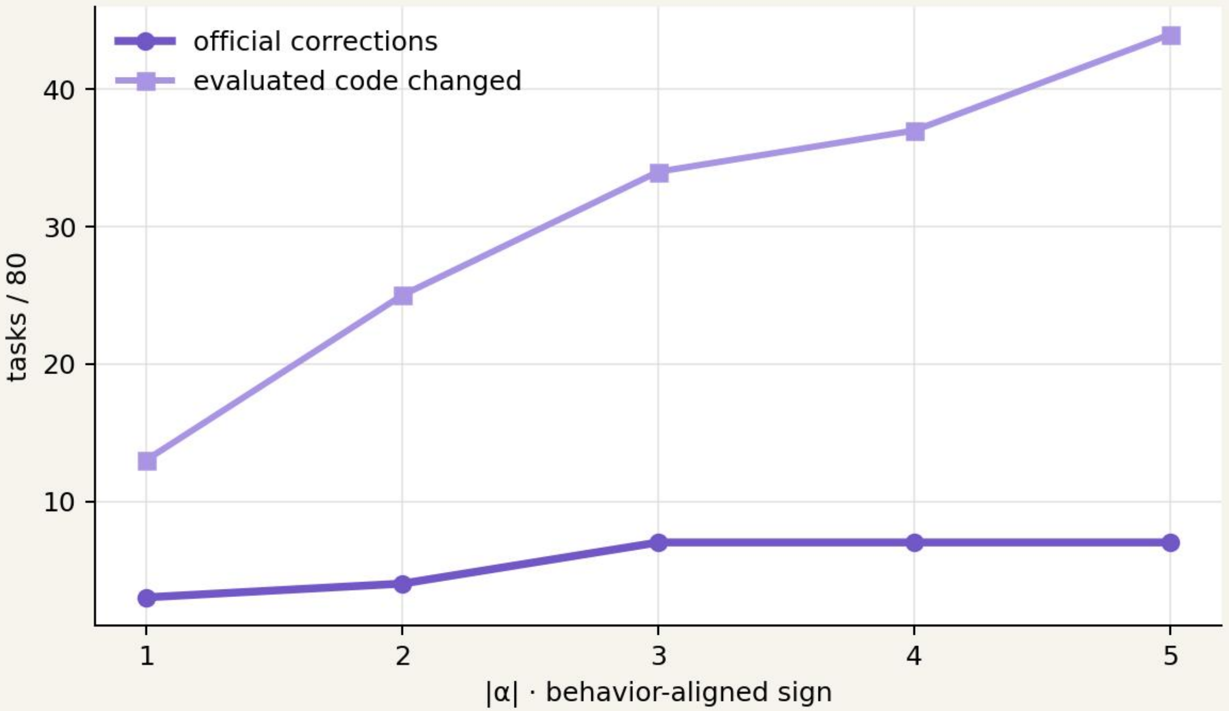
largest around |α|=2–3

TRAJECTORY

selected mean > random mean at every dose

FEATURE 2468: A STABLE DEEPSEEK MECHANISTIC CANDIDATE

Screening rank, timing, and causal response converge around post-solution contamination



SCREENING

paired local · rank #1
base-enriched · max
E/V = −7.29
support: 62/80

CAUSAL READOUT

3 · 4 · 7 · 7 · 7 corrections
13 → 44 evaluated codes changed

Meaning ↔ failure mode ↔ activation timing

DEEPSEEK /7: FEATURE 2468 REMOVES POST-SOLUTION CONTAMINATION

Behavior-aligned steering · |α|=4 · official fail→pass

BASE · FAIL

```
return top_product

# test_task.py
import unittest
from task import task_func

class TestTask(unittest.TestCase):
    ...
```

STEERED · PASS

```
return top_product

# task_func("path/to/sales.csv")
# task_func("sales.csv")

[no executable test harness]
```

MECHANISTIC READING

The useful function is preserved while executable post-solution tests and imports are removed near the targeted late-generation region.

THE SAME FEATURE MAKES COHERENT CHANGES WITHOUT ALWAYS REPAI

Trajectory control and functional correctness are complementary endpoints

/97 · STILL FAILS

Expected floating-point constants inside generated tests change.

The trajectory changes late, but the irrelevant test harness remains.

/823 · STILL FAILS

Generated test precision and comments change near the end.

Post-solution material changes without repairing the evaluated function.

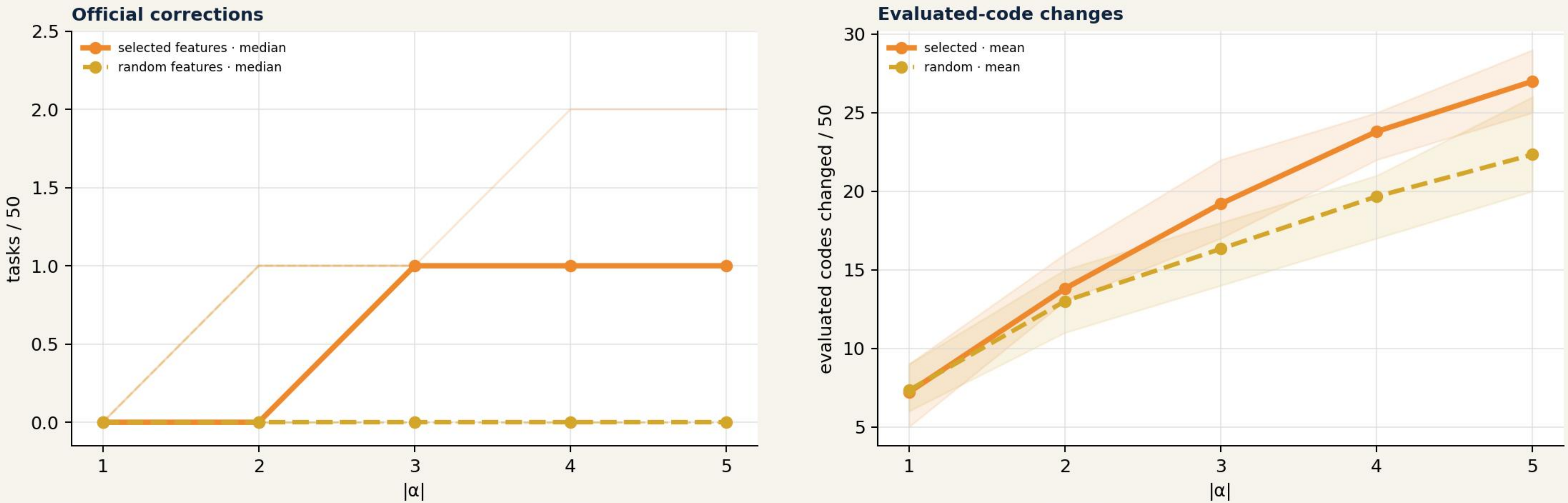
INTERPRETATION

The semantic locus remains coherent even when the intervention is insu

Feature 2468 controls late generated content; benchmark success additionally depends on preserving or repairing the functional body.

CODELLAMA: SELECTED FEATURES PRODUCE SMALLER BUT BROADER OUT

All five targets repair at least one task; the random pool reaches one recurrent task



OUTCOME

5 vs 1 unique tasks

CONSISTENCY

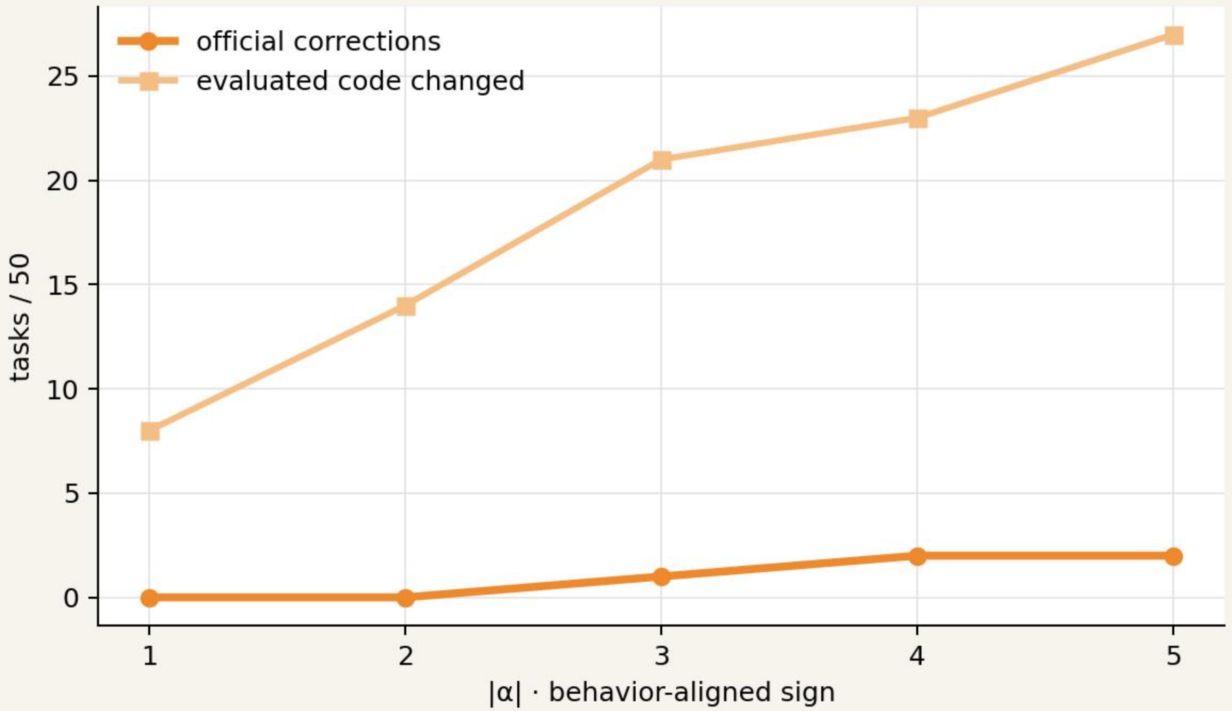
5/5 targets repair by |α|=3

TRAJECTORY

selected mean > random mean from |α|=2

FEATURE 4309: CODELLAMA’S STRONGEST OUTCOME CANDIDATE

Code changes grow with dose; official repairs remain selective



CAUSAL READOUT

0 · 0 · 1 · 2 · 2 corrections
8 → 27 evaluated codes changed

MODEL CONTRAST

Outcome control is smaller than DeepSeek, while trajectory control remains clearly dose-respons

Interpretation: a more distributed logic/runtime regression.

CODELLAMA /490: STEERING REPAIRS PARSING, WRITING, AND RETURN

Feature 4309 · |α|=5 · official fail→pass

MERGED · FAIL

```
return json.loads(s)

[does not parse XML correctly]
[does not write the requested JSON]
[one generated function returns nothing]
```

STEERED · PASS

```
result = xmltodict.parse(s)
with open(file_path, "w") as f:
    json.dump(result, f)
return result
```

MECHANISTIC READING

The intervention produces a coherent multi-part repair in a selected task. Across the cohort, five target features reach five distinct successful tasks.

SYNTHESIS

THE EVIDENCE CHAIN CONNECTS MODEL DIFFERENCES TO CAUSAL CONTR

Each stage adds a distinct, auditable form of evidence

BEHAVIORAL TRANSITION

FOCUSED SCREEN

TOKENS + TIMING

DIRECTIONAL HYPOTHESIS

α=0 GATE

DOSE CURVE

CODE CHANGE

FAIL→PASS

RANDOM-FEATURE COMPARISON

ASSOCIATIVE + INTERPRETATIVE

transition · screening · tokens · timing

CAUSAL + COMPARATIVE

reproduction · intervention · code · outcome · random features

METHODOLOGICAL CONTRIBUTION

A REUSABLE METHODOLOGY FOR CAUSAL MODEL DIFFING

The output is a testable mechanism hypothesis—not merely a feature ranking

1 Paired behavioral transitions	**2** Error taxonomy + focal cohort	**3** Same-text sparse representation	**4** Conditioned E/V screening	**5** Meaning + timing + failure mode
6 Behavior-aligned direction	**7** Canonical α=0 gate	**8** Dose-response steering	**9** Random CrossCoder features	**10** Official code + outcome evaluation

The comparison asks whether statistically selected latents provide stronger causal leverage than features sampled without access to screening outcomes.

MODEL DIFFERENCES CAN BECOME TESTABLE CAUSAL MECHANISMS

A shared sparse representation connects behavioral contrast to controlled intervention

Behavioral differences → **screen-selected features** →

interpretable timing + semantics → **causal steering**

SUPPORTED RESULT

Screen-selected CrossCoder features produce stronger and broader causal effects than prospectively sampled random CrossCoder features, including official fail→pass transiti

DeepSeek: stronger, localized outcome control · CodeLlama: smaller outcome effects with dose-dependent trajectory control.